1.1 Condition Coverage Table

Test ID	C1 (no patient)	C2 (past time)	C3 (weekend)	C4 (bad hour)	C5 (bad type)	C6 (bad dur)	Expected Outcome
TC1	T	-	-	-	-	-	ValidationError - Patient not found
TC2	F	T	-	-	-	-	ValidationError - Appointment must be in the future
TC3	F	F	T	NotEvaluated	-	-	ValidationError - Weekend appointment
TC4	F	F	F	T	-	-	ValidationError - Outside clinic hours
TC5	F	F	F	F	T	-	ValidationError - Invalid appointment type
TC6	F	F	F	F	F	T	ValidationError - Invalid duration
TC7	F	F	F	F	F	F	Appointment created successfully

1.2 - Short-Circuit Evaluation

a) When C3 is True, the appointment is on a weekend. Because Python evaluates OR expressions from left to right, the whole condition is already True once C3 is True. Therefore C4, the bad-hour check, is not evaluated. This means appt_dt.hour is not accessed when the appointment date is a weekend and it immediately raises a ValidationError for invalid appointment time.

b) Test:

```
test_hw2_conditions.py 3 ×
ers > bibekdhakal > Desktop > MediTrack_ProjectNew > meditrack_src > tests > test_hw2_conditions.py > ...
def test_reschedule_success(sample_patient, future_monday_10am):
    appt_id = appointments.schedule_appointment(
        sample_patient, "provider-001", future_monday_10am, "follow_up", 30
    )
    appointments.cancel_appointment(appt_id, "staff", "reason")

    new_id = appointments.reschedule_appointment(
        appt_id, future_monday_10am + timedelta(days=1), User("admin")
    )

    assert appointments.get_appointment(new_id)["status"] == "Scheduled"


def test_short_circuit_weekend(sample_patient):
    fake_dt = MagicMock()
    fake_dt.__le__.return_value = False
    fake_dt.weekday.return_value = 6

    type(fake_dt).hour = property(
        lambda self: pytest.fail("hour should not be evaluated")
    )

    with pytest.raises(ValidationError):
        appointments.schedule_appointment(
            sample_patient,
            "provider-001",
            fake_dt,
            "follow_up",
            30
        )
```

Verification Result:

```
bibekdhakal@Bibeks-MacBook-Air meditrack_src % pytest tests/test_hw2_conditions.py -v
=========================================================================================== test session starts ==================
platform darwin -- Python 3.14.5, pytest-9.0.3, pluggy-1.6.0 -- /Library/Frameworks/Python.framework/Versions/3.14/bin/python3.14
cachedir: .pytest_cache
rootdir: /Users/bibekdhakal/Desktop/MediTrack_ProjectNew/meditrack_src
plugins: cov-7.1.0, Faker-40.19.1
collected 19 items

tests/test_hw2_conditions.py::test_patient_not_found PASSED
tests/test_hw2_conditions.py::test_time_in_past PASSED
tests/test_hw2_conditions.py::test_invalid_type PASSED
tests/test_hw2_conditions.py::test_invalid_duration PASSED
tests/test_hw2_conditions.py::test_valid_appointment PASSED
tests/test_hw2_conditions.py::test_weekend_rejected PASSED
tests/test_hw2_conditions.py::test_outside_hours_rejected PASSED
tests/test_hw2_conditions.py::test_cancel_missing_appointment PASSED
tests/test_hw2_conditions.py::test_check_in_wrong_state PASSED
tests/test_hw2_conditions.py::test_complete_wrong_provider PASSED
tests/test_hw2_conditions.py::test_start_visit_missing PASSED
tests/test_hw2_conditions.py::test_start_visit_wrong_state PASSED
tests/test_hw2_conditions.py::test_complete_missing PASSED
tests/test_hw2_conditions.py::test_complete_empty_notes PASSED
tests/test_hw2_conditions.py::test_reschedule_missing PASSED
tests/test_hw2_conditions.py::test_reschedule_wrong_state PASSED
tests/test_hw2_conditions.py::test_reschedule_unauthorized PASSED
tests/test_hw2_conditions.py::test_reschedule_success PASSED
tests/test_hw2_conditions.py::test_short_circuit_weekend PASSED

=========================================================================================== 19 passed in 0.04s ==================
bibekdhakal@Bibeks-MacBook-Air meditrack_src %
```

The test test_short_circuit_weekend PASSED and the entire test_hw2_conditions.py reported 19 passed tests. Since the test would fail immediately if appt_dt.hour were accessed, the passing result proves that C4 was not evaluated when C3 was True.

2.1 - Design 5 Loop Tests

Test ID	Loop Iterations	Overlap?	Expected Outcome	Jorgensen Case
LT1	0	No	Appointment scheduled	Zero iterations
LT2	1	No	Appointment scheduled	One iteration (no overlap)
LT3	1	Yes	SchedulingConflict raised	One iteration (with overlap)
LT4	N - 1 = 7	No	Appointment scheduled	N-1 iterations
LT5	N = 8	No	Appointment scheduled	N iterations

2.2 - Coverage.py Report

Coverage Report :

Branch coverage was collected for appointments.py using Coverage.py while executing the loop tests in tests/test_hw2_loops.py.

Commands used:

```
coverage run --branch -m pytest tests/test_hw2_loops.py
coverage report -m --include=appointments.py
```

Coverage Result:

- File: appointments.py
- Branch Coverage: 95%

The achieved branch coverage passes the required target of 75%.

Coverage Result :

A screenshot of the Coverage.py report is included below:

```
bibekdhakal@Bibeks-MacBook-Air meditrack_src % coverage run --branch -m pytest tests/test_hw2_loops.py
coverage report -m --include=appointments.py
=========================== test session starts ===========================
platform darwin -- Python 3.14.5, pytest-9.0.3, pluggy-1.6.0
rootdir: /Users/bibekdhakal/Desktop/MediTrack_ProjectNew/meditrack_src
plugins: cov-7.1.0, Faker-40.19.1
collected 26 items

tests/test_hw2_loops.py ..........................                  [100%]

=========================== 26 passed in 0.08s ============================
Name              Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------------
appointments.py      89      4     48      3    95%   82, 90, 148-152
-------------------------------------------------------------
TOTAL                89      4     48      3    95%
bibekdhakal@Bibeks-MacBook-Air meditrack_src %
```

3.1 - Switch Coverage Table

Test ID	Invoice.status	amount vs balance	Expected Outcome
SW1	None (not found)	-	ValidationError - Invoice not found
SW2	"Draft"	-	InvalidStateError - Submit invoice first
SW3	"Voided"	-	InvalidStateError - Cannot pay voided invoice
SW4	"Paid"	-	InvalidStateError - Already fully paid
SW5	"Submitted"	amount > balance	ValidationError - Payment exceeds balance due
SW6	"Submitted"	amount == balance	Payment accepted; invoice status becomes Paid
SW7	"Submitted"	amount < balance	Payment accepted; invoice status becomes PartiallyPaid
SW8	"PartiallyPaid"	amount == balance	Payment accepted; invoice status becomes Paid
SW9	"unknown_status"	-	ValueError - Unknown invoice status

4 – Full Coverage Analysis

Coverage Results:

appointments.py - 97%

billing.py - 93%

Both modules gets to the required coverage targets.

Uncovered Lines and Branches:

1. appointments.py (lines 82 and 90): These lines correspond to defensive validation branches that require unusual appointment states and are difficult to reach through normal scheduling operations.

2. billing.py (lines 33–40): These lines belong to an internal billing validation path that requires specific invoice configurations which rarely occur during normal workflows.

3. billing.py (lines 45 and 137–138): These branches represent uncommon invoice-processing scenarios and special error conditions that are difficult to reproduce in standard test cases.

Coverage Report:

```
bibekdhakal@Bibeks-MacBook-Air meditrack_src % coverage report -m
Name                              Stmts   Miss Branch BrPart  Cover   Missing
--------------------------------------------------------------------------
appointments.py                      89      2     48      2    97%   82, 90
billing.py                          102      7     48      2    93%   33-40, 45->44, 137-138
config.py                            23      0      0      0   100%
exceptions.py                        14      0      0      0   100%
models.py                            21      8      6      0    48%   36-42, 50-51, 56
patients.py                          81     42     48     12    40%   15, 17, 20, 25, 28, 33, 35, 40, 43, 46, 62-66, 72-76, 101-115, 120-129
prescriptions.py                     67     55     36      0    12%   20-23, 27-30, 72-129, 138, 151-158, 168-176, 193-206
records.py                           60     47     32      0    14%   25-28, 35-39, 73-116, 121, 126-129, 141-145, 162-175
tests/__init__.py                     0      0      0      0   100%
tests/conftest.py                    73     17      0      0    77%   55, 60, 65, 70, 89, 106, 111, 122-130, 135, 147, 159-160, 165
tests/test_hw2_conditions.py         87      0      0      0   100%
tests/test_hw2_loops.py             119      0      4      0   100%
tests/test_hw2_switch.py            133      1      2      1    99%   159
--------------------------------------------------------------------------
TOTAL                               869    179    224     17    73%
bibekdhakal@Bibeks-MacBook-Air meditrack_src %
```